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Atomization processes of metal powders for 3D printing

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ABSTRACT

Additive manufacturing also known as 3D printing becomes popular over the past decades. The process of additive manufacturing uses layer by layer deposition of material and produces products directly from 3Dmodel. The considered technology has capability for manufacturing product with complexity and with high precision in short time. Therefore, additive manufacturing is widely used in industries and biomedicine. In case of metal 3D printing, metal powders are used as main material to produce the product. There are several methods for metal powder production and one of the popular methods is atomization method. The main aim of this study was to review existing processes of atomization method, compare the costs and application purposes. The four main processes known as "gas atomization", "water atomization", "centrifugal atomization" and "plasma atomization" were taken into consideration. The comparison of atomization processes has demonstrated that plasma atomization provides highly spherical shape powders however it is considered expensive process, while the gas atomization produces spherical powders with optimal cost and appropriate shape. In addition, powders obtained from plasma atomization is the best for biomedical application.

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1. Introduction

During the fabrication of product, conventional manufacturing tools remove unwanted material from the work piece in subtractive processes, while 3D printing/additive manufacturing (AM) processes are developing product via joining material layer by layer [1]. These AM processes are capable to create complex and various parts, and have the following advantages:

- Opportunity to manufacture parts with complex geometries and internal features.
- Suitable for small batch sizes as it can directly go from 3d digital model to manufacturing.
- There is no need in path planning and tooling that make AM customization quick and inexpensive.
- AM can manufacture hard metals which is hard for traditional removal processes [2].

It is not difficult to foresee the prospective of the AM impact [3], while there is a hype around the vision of AM. However, there is a high potential of the AM in every industry that it could affect [4]. The application of metal powder processing includes itself production of commercial and developmental materials for industries. There exists different techniques of metal powder development: gas atomization, water atomization, centrifugal atomization, plasma atomization, mechanical attrition and alloying, melt spinning, rotating electrode process (REP), and a variety of chemical processes [5]. Each of listed processes produces metal powders that are different in their morphology, shape and size. Almost all processes create spherical shape powders with irregular surface [6]. The analysis of AM material focuses on its physical properties {ductility and hardness}, chemical properties {impurities and reactivity}, bulk properties {tap density, apparent density, compressibility, green strength, flow properties and compressibility} [5]. Generally, high quality metal powders with fine size distribution, and without oxide contaminants are used to produce materials for structural and functional application. Economically, the metal powder production and AM are better than the traditional forging and casting processes [7]. The following scheme in Fig. 1 shows processes available for mass powder production. This paper is focused on atomization mechanical methods.

The spherical shape and narrow size distribution of metal powders are major requirements of AM for obtaining repeatable and reliable results. The technologies collected in this review allow to

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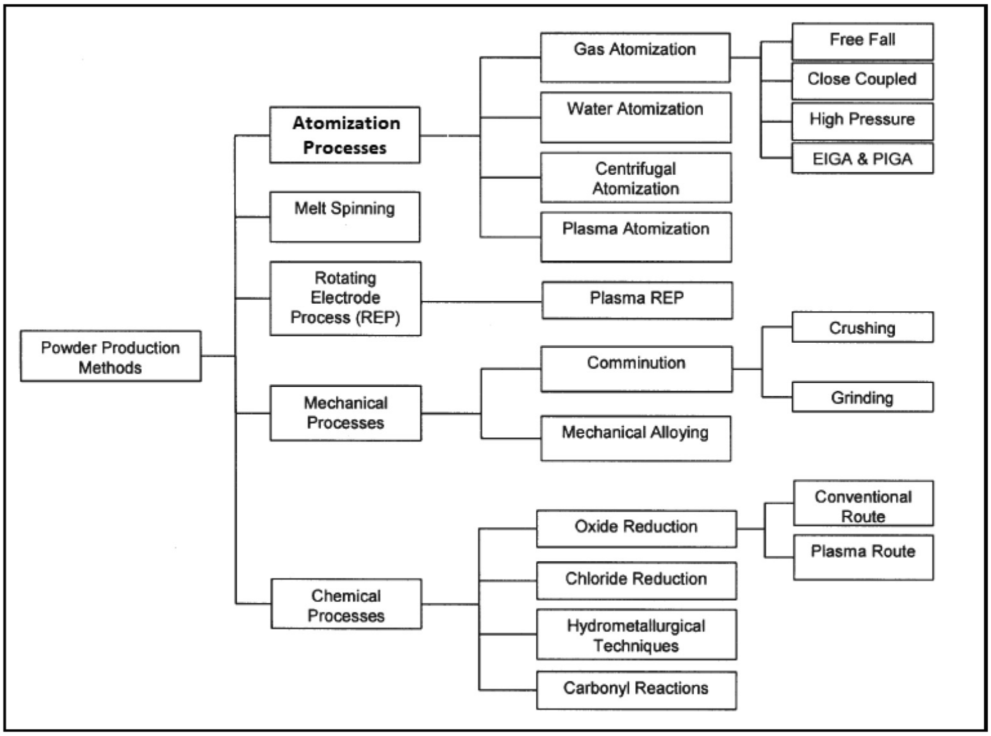

Fig. 1. Powder production methods and processes [5,8].

produce powder shape that are spherical or near to spherical form. Moreover, the results are achieved right after the powder synthesis, while the other processes need additional processing to achieve the desired powder form [10]. There are two main methods for metal powder production: mechanical and physical-chemical methods. The change of the workpiece via chemical and physical transformations of raw material is associated with the physical–chemical methods of powder production. The jet dispersion melts under high pressure of fluid or different types of mechanical milling processes which results in a powder production associated with the mechanical methods [11]. This review is focused on the atomization mechanical methods of the powder production.

The Fig. 2 below shows the general system of atomization powder production that consist of three stages. In the stage 1 preprocessing takes place, where metal is melted, and output gives feedstock for atomization process. During the stage 2, fluid metal is passing through the (gas, water or centrifugal) atomization process, before post processing of the powder and its validation [9].

The main aim of this paper is to overview one of the popular mechanical method of powder production for metal 3D printing. Besides, the review has detailed description of popular atomization processes from the feedstock selection to the metal powder processing.

2. 1.1Feedstock selection

The initial feedstock material parameters such as geometry and shape affect the melting process, and, they also affect the quality of the produced powder (Table 1). The melting technology with application of crucible can take materials such as pure metal elements, powder, scrap, metal alloy elements or pre-alloyed. Generally, such melting system installs sample taking device to get liquid melt [12]. Such sampler helps to obtain test of the molten metal and observes the chemistry for analysis. In some cases, this sample analysis results show that process requires additional alloying elements. The various kind of raw materials can be combined using

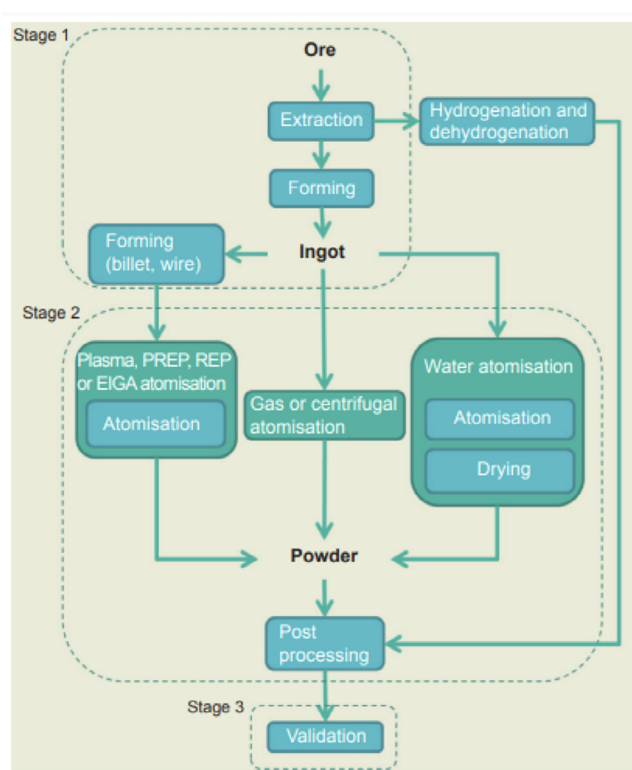

Fig. 2. Schematic illustration of the atomization process [9].

Table 1
Feedstock types used in manufacturing.

Type of feedstock	Scrap, Pre-alloyed, Alloyed...etc.
Chemical composition	Aluminum, Chromium, Nickel, Cobalt...etc.
History of the feedstock	Remelted bar, Drawn wire, Powder...etc.

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that furnace. If there is no sample taking device in the furnace, the homogeneous raw metals with the same chemistry must be used for desired metal powder production [13].

Some methods of powder metallurgical processing require the definite geometry of the feedstock. For instance, plasma atomization process requires metal powders or metal coil of wire as a feedstock. For titanium powders, the cast electrodes are mainly used as a feedstock [14].

2.1. Melting of feedstock

Mainly, the melting process is accomplished via open or closed melting systems. During the open melting systems, which are always done on open air, the layer of liquid slag covers the liquid melt. The most metallurgical operations are done via slag in the open melting [15]. The closed melting systems use vacuum as a melting environment which has several benefits:

- Safe raw material degassing without air pollution
- There is no oxidation of reactive material elements
- Exclusion of dissolved gases from the system, e. g. oxygen, nitrogen, hydrogen
- Exclusion of undesired trace material elements from the system

The size of the batch in closed melting system is usually smaller compared with the open furnace. Moreover, the metal powders for additive manufacturing are mostly developed by applying inductive melting technology which is accomplished in vacuum. According to the type of the feedstock, the systems of metal melting technologies can be split into liquid melt pool system and partially molten feedstock system [14,15]. The main charge source of the melt pool is a crucible with inductive heating. For melting the reactive materials such as titanium (Ti), the water-cooled crucible is appropriate.

3. Atomization processes

Consequently, the next process in AM after melting is atomization [5] that consists of two steps. First of all, there is a shear of liquid metal, and next step is rapid cooling of metal droplets. Initially, a large volume of metal melt is split into smaller metal droplets. The melt is shearing due to the high-velocity gas/water/plasma jets or centrifugal operations. Therefore, atomization processes respectively divided into gas/water/plasma atomization and centrifugal. During the next step, the shape of the droplets starts to develop, and spherical droplets start to solidify. The temperature of the heated melt droplets and surface tension of the melted material influence to the speed and efficiency of the process [8].

3.1. Gas atomization

Currently, gas atomization is the leading technology for producing metal powders for AM. The feedstock material has three process steps such as melting, atomization and solidification that are included into the production management. Every gas atomizer contain furnace for metal melting under protective atmosphere or under vacuum conditions. The thin flow of melted liquid metal dispersed into small droplets under the high pressure of gas. During the fall, all droplets solidify before reaching to the chamber. Usually, fine and spherical shape powders are developed by gas atomization. It should be noted that particle size distribution has a strong dependence on the type of atomized alloy and used system [20].

3.1.1. Free fall atomization

In this branch of gas atomization technique, the liquid metal stream is falling from the crucible to the certain distance before meeting with gas jets at some point. It can be seen from the Fig. 3a [16]. The molten metal stream turns to the spherical particles with different diameters and is solidified after passing the point of collision. The shower of droplets appear after that point. The results of the process show log normal size distribution of metal powders. To reduce the median mass size of particles, the gas and metal flow velocity should be increased, and this is valid for the angles of attack between the fluid flow and liquid metal streams [16,17].

3.1.2. Close-coupled gas atomization(CCA)

The close coupled (confined) atomization CCA is the most popular process among the gas atomization technologies [18]. Main difference between free fall and close coupled gas atomization is in the position of the gas jet (Fig. 3b). In CCA the gas delivery tubes placed just under the exit of the melt feeding. The distance of gas jet tube in free-fall atomization is fluctuating from (10–13) cm to 30 cm. In CCA processes, there is no long movement of melt stream directed downward in quiescent air. In that moment, the stream immediately disrupts the flow of liquid metal. The advantage of the CCA is its capability of producing fine powders with smaller configuration. The disadvantages of CCA is backflow and freeze off. To achieve a better dispersion of the liquid phase, the distance between metal melt and jet of gas must be shortened [17,18].

3.2. EIGA and PIGA processes

The presence of contamination of ceramic pieces produced by the lining of the induction furnace and tundish in the final product was one of the issues of using gas atomization methods [19]. Consequently, this problem led to the discovery of new techniques as EIGA and PIGA. Both introduced techniques do not require ceramic liner material to create clean reactive metal powders. To perform the EIGA, prefabricated electrodes of desired material are melted at the front-end section and filled into the nozzle system, where this mixture is atomized by a high-velocity gas stream as illustrated in Fig. 4.

During the PIGA process, specific pressure is required to atomize the desired material.

The difference between those processes lies in using the different heat source to melt the material. Specifically, In EIGA prefabricated electrodes are melted using inductive coils while in PIGA metal is melted by plasma gun. As a result, a high quality inert gas atomization is obtained using these techniques.

3.3. Water atomization

The following scheme (Fig. 5) shows the process of water atomization of the melted metal. This process uses gas jet and water jet directed to the molten metal. Initially, material placed into the induction furnace and is melted before pouring to the tundish. In the next stage, liquid metal experiences fine jet of gas and high-pressure pump blows water jet as shown in the Fig. 5 until the material becomes atomized. The resulting mix of water and metal powder proceeds to the drying and sorting according to the desired sizes. The water atomization nozzle (V-jet nozzle), designed by Kobe Steel [9], increases atomization ability of the process. Prevention of melt stream oxidation can be done by doing the melting operation in an inert atmosphere or with installation of inert gas filled chamber around the water jet nozzle [11]. The optimum level jet pressure gives a high-quality powder. If there will be deviation of the pressure, the atomization will be with irregularities and results in high particle size deviation [4,9].

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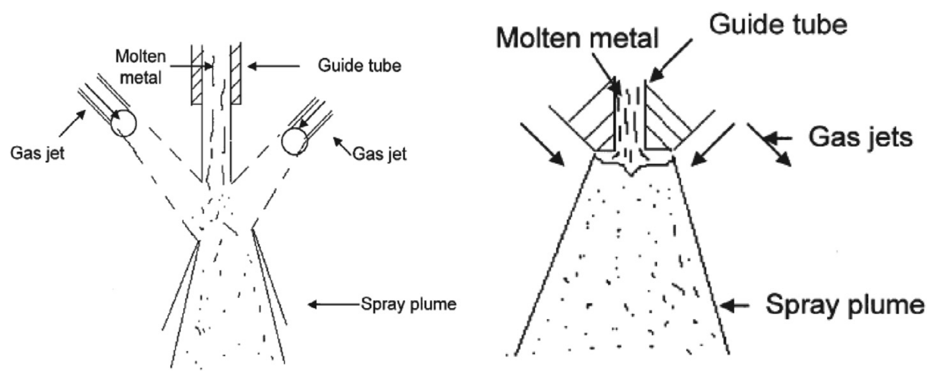

Fig. 3. (a) The process of free fall gas atomization [17] (b) The process of close-coupled gas atomization [17].

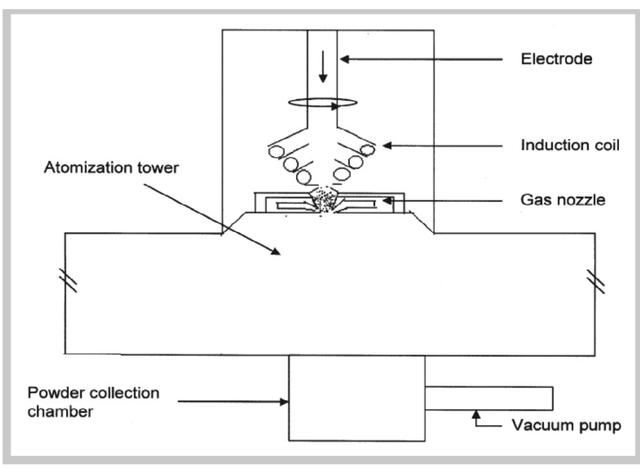

Fig. 4. A schematic sketch of an EIGA process [7].

3.4. Plasma atomization

Atomization in this process includes application of plasma in order to produce powders of high purity. It is relatively new and more advanced. The process can produce powders with highly spherical particles and low oxygen content [20,21]. The material used in plasma atomization comes in the form of metal wire. It is fed into plasma torches, which transforms it into droplets that are later solidified into powder form. Powder produced by plasma atomization has particle size distribution of 0 to 200 µm [22,24]. The process of plasma atomization has shown in Fig. 6.

In comparison to their advantages, plasma atomization has a list of significant disadvantages. The first one is that initial material should be able to transform into wire form, meaning that materials that cannot be in that form, are impossible to be atomized by plasma [25]. Two companies, AP&C (ex Raymor) and PyroGenesis, use plasma atomization as the main process for powder production. Both companies are focused on applications of powder in additive manufacturing [23]. The main advantage of wire usage in plasma atomization over gas atomization is that the metal feedstock and its melted form does not encounter low temperature solid surfaces. Powders produced by plasma atomization are

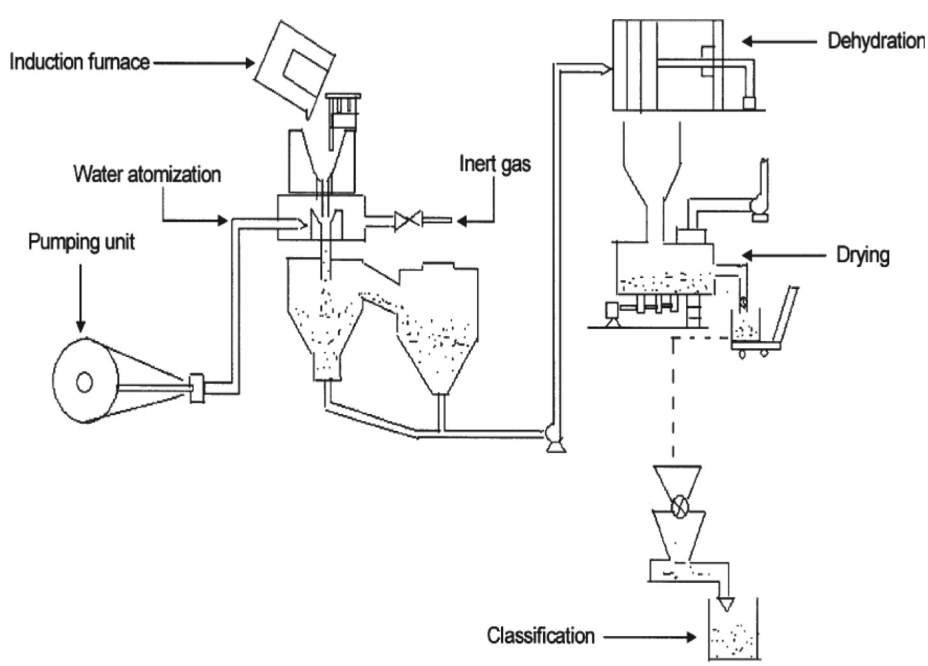

Fig. 5. A schematic sketch of water atomization [5,36].

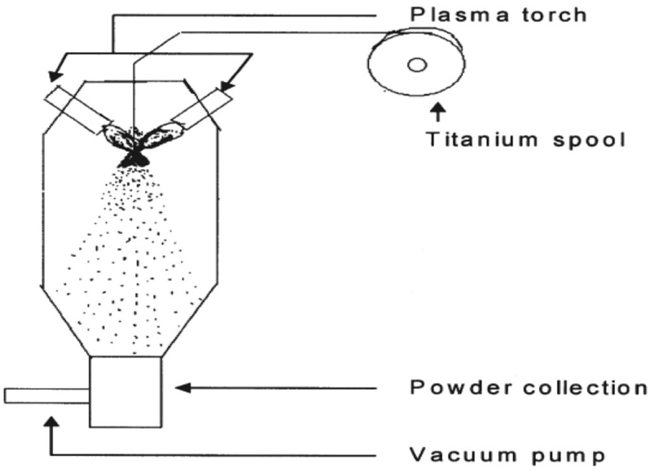

Fig. 6. Plasma atomization [41,42].

widely used in biomedicine and in sporting goods [26–29]. The companies named Pyro-Genesis and Hydro-Quebec (LTEE) are focused on developing titanium powders in different size range and becomes a patent owner of the plasma atomization process. The same technology are used to produce powders from copper, molybdenum and nickel-based super-alloy. The process takes place in an inert atmosphere along with high velocity of hot ionized jet of inert gas which produces proper atomization. Since the process is going inside the inert atmosphere, the surface structure of powders has insignificant level of impurities compared to other mentioned processes [26–29].

3.5. Centrifugal atomization

Centrifugal atomization, as it can be predicted uses the power of the centrifuge, which shears the molten metal in a high-speed, thus creating a fine metal powder (Fig. 7). The powder produced by this method has its mean diameter of above 100 μm. The reason is that centrifugal atomization is restricted due to mechanical friction that has its effect on the speed of the rotor which controls the average particle size [32]. Development of self-lubricating centrifugal atomization equipment for fluid spray rotation, increases the rotor speed 5 to 6 times and allows to achieve finer particles of powder. Furthermore, combining the previous technique with the gas atomization's spray powder technique helps to produce Al-20Si alloy of finer size of 7–8 μm. Here, superheated molten metal passes through funnel, the jet then the leaf-type rotary plate. This results in molten metal being atomized twice. The combined technique has higher working efficiency, high speed of the rotary plate and convenient speed regulation (Fig. 8) [30–34].

3.6. Discussion

The table shown below list of noted atomization techniques, shape and size of powder particles, advantages, disadvantages of techniques and commonly used materials for production. According to the table, it is seen that particle size of processes varies, and if maximum particle size of the powders can be produced with centrifugal atomization process, the plasma atomization can produce the smallest powder particles [38]. Among the discussed processes, the most spherical shape obtained from plasma atomization. However, this method is the most expensive and requires only wire or powder feedstock, while the gas and water atomization are relatively less expensive production process.

Fig. 9 presents the images of the various powder particle shapes obtained from the various processes [36–38]. Besides, this figure justifies the data given in the Table 2.

4. Powder processing

At the final step of the powder development, the produced metal powders diameters vary between 0 and 250 μm and it depends on the used methodology. Application of produced powder in metallic 3D printing require fraction separation of the particles. Lastly, air classification methods or sieving can be realized to do sorting of the powder. Before releasing the product, the properties of metal powders must be measured and checked in terms of all properties mentioned in introduction. Currently, the powder material characterization and various procedures for investigation are available in the literature [18,40].

5. Discussion and conclusion

From the literature reviewed during the preparation of this script, it was noticed that the application of metal powders in the industries for producing complex geometry parts is growing rapidly over the last decades. Various powder production technologies for metallic powders have been established. Besides, the significant growth of 3D printed components have created high demand of metal powder for the AM printers. With the increase

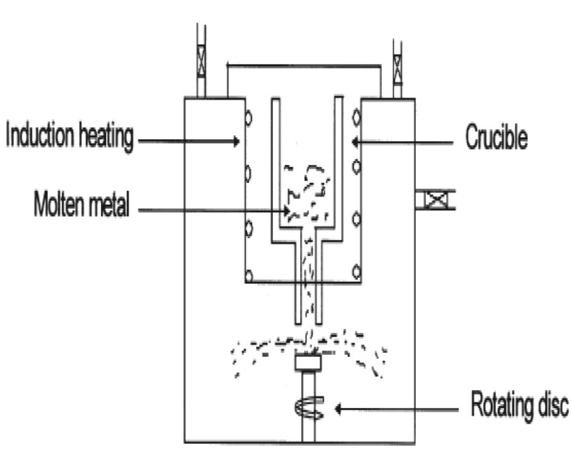

Fig. 7. Manufacturing process of rotating-disk [43].

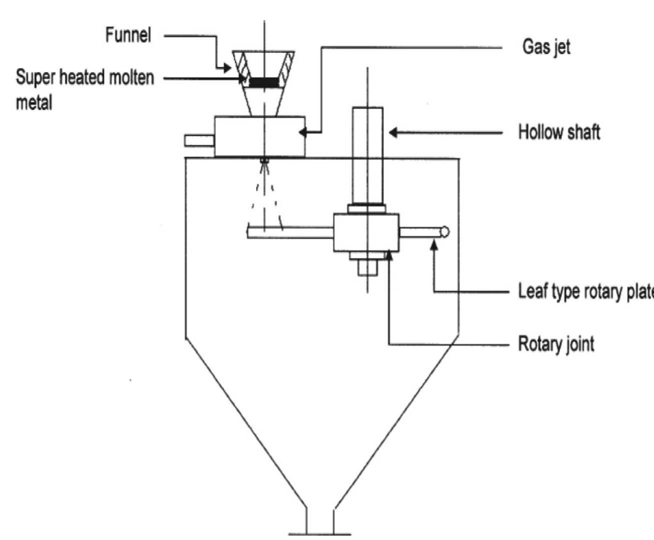

Fig. 8. Centrifugal atomization with gas jet [44].

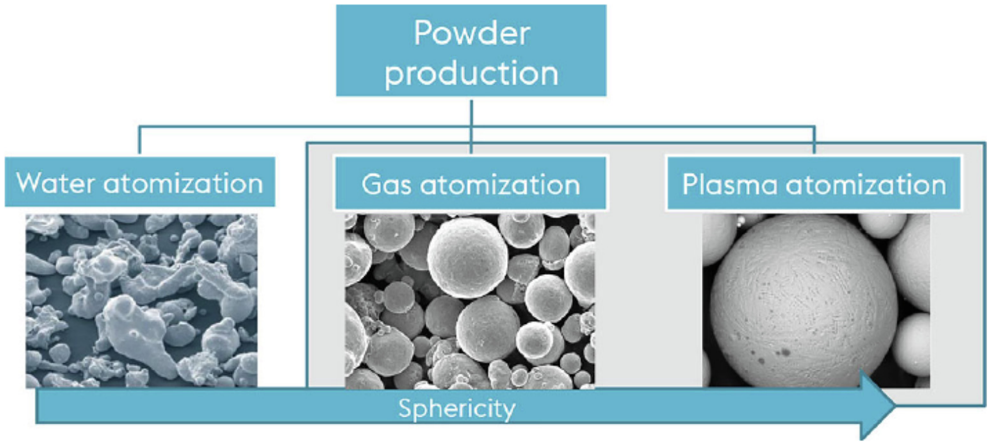

Fig. 9. Shapes of different metal powders [15].

Table 2
Metal particle characteristics by manufacturing methods [9,19,35,37–40].

Manufacturing process	Particle size, μm	Particle shape	Advantages	Disadvantages	Common use
Gas atomization	0–500	Spherical shape	High throughput Range of particle sizes Only requires feedstock in ingot form	Post processing required to remove water Irregular particle morphology Satellites present Wide PSD Low yield of powder between 20 and 150 μm	Non-reactive
Water atomization	0–500	Irregular droplet shape	Wide range of alloys available Suitable for reactive alloys Only requires feedstock in ingot form High throughput Range of particle sizes Use of EIGA allows for reactive powders to be processed Spherical particles	Satellites present Wide PSD Low yield of powder between 20 and 150 μm	Ni, Co, Fe, Ti (EIGA), Al
Plasma atomization	0–200	Highly spherical shape	Extremely spherical particles	Requires feedstock to either be in wire form or powder form High cost	Ti (Ti64 most common)
Centrifugal atomization	0–600	Spherical shape	Wide range of particle sizes with very narrow PSD	Difficult to make extremely fine powder unless very high speed can be achieved	Solder pastes, Zinc of alkaline batteries, Ti and steel shot

of the demand, the optimization of the processes and increased product quality should be the target of the manufacturers. The metal powders are the key component in metallic AM and becomes the most important in 3D printing. For that purposes, this paper briefly described the different methods of powder production and mainly focused on the atomization processes which refers to the mechanical method. Generally, all sections in the main part approaches to thoroughly describe the processes of the metal powder production and individual feature of each technology. After the overview, the table of main particle characteristics were presented. From the Table 2, it was concluded that the plasma atomization technology can provide highly spherical shape of the particles with the relatively smooth surface structure. However, the cost of this method is the highest and it could be replaced with the other atomization processes according to the customer requirements.

CRediT authorship contribution statement

Kazybek Kassym: Conceptualization, Investigation, Methodology, Writing - original draft. **Asma Perveen:** Visualization, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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